BSc (Hons) in **Information Technology Specialized in AI**

Faculty of Computing

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Practical 04

> **Objective & Lecture Mapping:** In previous labs, you explored Uninformed Search strategies like BFS and UCS inside `agent.py` [cite: 1]. Today, we transition to **Informed Search**. You will enhance your agent by implementing the **A* Search** algorithm, using **Heuristics** to drastically reduce the number of explored nodes in the `visual_grid_game.py` environment[cite: 4]. You will start with basic heuristic functions and connect them to your search logic.

Part 1: Practical Implementation — Building the A* Agent

Step 1.1: Implementing the Heuristic Functions

> Informed search algorithms require a heuristic function, h(n), which estimates the cheapest path from the current node to the goal. You will calculate distance metrics on a 2D grid.

1. Create a new Branch called Week_04 in your Forked Github Repo.
2. Open `agent.py` and locate your `SearchAgent` class[cite: 1].
3. Add a new method named `def manhattan_distance(self, pos, goal):` that calculates the distance using the formula h(n) = |x_1 - x_2| + |y_1 - y_2| and returns the integer value.
4. Add a second method named `def euclidean_distance(self, pos, goal):` that calculates the straight-line distance using the formula h(n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}. You will need to `import math` for this.
5. *Testing Checkpoint:* Print the outputs of both functions for a mock start position (0, 0) and goal (3, 4) to verify that Manhattan returns 7 and Euclidean returns 5.0.

Step 1.2: Implementing A* Search

> A* evaluates nodes by combining the path cost so far, g(n), and the estimated cost to the goal, h(n). The evaluation function is f(n) = g(n) + h(n).

1. Inside `SearchAgent`, create a new method: `def astar_search(self, start_pos, goal_pos, walls, grid_size, heuristic_type='manhattan')`.
2. Initialize an empty priority queue using `heapq` and an empty set for `reached_states`.
3. Push the initial state into the priority queue. Unlike UCS which only uses g(n), your A* tuple must be formatted as: `(f_cost, g_cost, current_pos, path_taken)`. For the starting node, g(n) = 0. Calculate h(n) using your chosen heuristic method, and set f(n) = g(n) + h(n).
4. Create your standard `while` loop to process the queue. Pop the node with the lowest `f_cost`. If `current_pos == goal_pos`, return the `path_taken`. Otherwise, add `current_pos` to `reached_states`.
5. For node expansion, check the four adjacent cells (Up, Down, Left, Right). For each valid neighbor (not a wall, within bounds, and not reached): Calculate g_{new} = g_{current} + 1, h_{new} using your heuristic, and f_{new} = g_{new} + h_{new}. Push the new tuple to the priority queue.

Step 1.3: Integrating A* into the Agent's Decision Loop

> Finally, you need to connect your new search algorithm to the agent's perception and action loop so it can run in the visual environment.

1. Navigate to the `sense_and_act(self, percept)` method in your `SearchAgent`.
2. Add an `elif self.active_algo == 'AStar':` block to handle the new algorithm alongside your existing BFS/DFS/UCS.
3. Extract the necessary global state from the percept dictionary (e.g., `percept['remaining_food']`).
4. Find the closest food item to act as the `goal_pos`. Call your `astar_search` method and store the returned list of actions in `self.plan`.

5. Open `visual_grid_game.py` and modify the `GridGameGUI` initialization to inject your `SearchAgent()` instead of the `ModelBasedAgent()`. Run the simulation and observe the optimized pathfinding!

Part 2: Theoretical Evaluation

> Based on your implementation and Lecture 05, complete the following analytical questions.

1. (Understand) What is the key difference between how Uniform-Cost Search (UCS) and A* Search prioritize which node to explore next?

UCS chooses the node with the **lowest path cost so far,** `g(n)`. A* considers both the **cost so far,** `g(n)`, and the **estimated cost to the goal,** `h(n)`, using `f(n) = g(n) + h(n)`. Therefore, A* can use the heuristic to guide the search toward the goal.

2. (Analyze) In Step 1.1, you used Manhattan Distance. Why is Manhattan Distance considered an "admissible" heuristic for this specific 4-way movement grid, and what would happen to your A* algorithm if the heuristic was NOT admissible?

Manhattan Distance is admissible in a 4-way grid because it counts the minimum horizontal and vertical moves needed to reach the goal. Walls can only make the actual path longer, so Manhattan Distance never overestimates the true cost. If a heuristic overestimates the cost, A may choose a longer path and is no longer guaranteed to find the optimal path

3. (Evaluate) If we modified `visual_grid_game.py` to allow the agent to move diagonally (8-way movement), would Manhattan distance still be an admissible heuristic? Why or why not? Which metric should you switch to?

No, Manhattan Distance would **not be admissible** if diagonal movement is allowed. This is because diagonal movement can reach the goal in fewer moves than Manhattan Distance predicts.

For example, from `(0,0)` to `(3,3)`, Manhattan Distance is **6**, but with diagonal movement the agent can reach the goal in **3 moves**.

Therefore, Manhattan Distance would overestimate the actual cost. We should switch to **Euclidean Distance**, which is suitable for 8-way movement.

4. (Create) When targeting multiple food items simultaneously, calculating the distance to just the single closest food item is a weak heuristic. Propose (in text) a stronger heuristic for navigating the grid to eat ALL remaining food efficiently.

Instead of considering only the distance to the nearest food item, we can use a **Minimum Spanning Tree (MST)** to estimate the cost of connecting all remaining food items. The heuristic can combine the distance from Pacman to the food network with the MST cost between the remaining food items. This gives a better estimate of the total cost required to collect all food items.

Once Completed Get a PDF of the completed File and Push into the Week 04 branch in the Github Project

Finalize and Lock Lab 04 Documentation